

Subject Description Form

Subject Code	ITC4072T
Subject Title	Colour Management and Technology
Credit Value	3
Level	4
Pre-requisite(s) / <Co-requisite> / (Exclusion)	ITC2203T Foundations of Textiles II
Objectives	The subject provides in-depth knowledge in colour management and technology required in the processes of colour measurement and colour quality control. It also examines the principles in colour communication, with emphasis on the hands-on experience of computer colorant formulation and colouration processes.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - explain the fundamental principles of colour, its perception and measurement; - describe the fundamental principles and instruments used in measuring and controlling the colour of textile material as well as the various factors which can affect the values so obtained; - describe the methods and standards used to specify colour internationally and apply them to communicate colour information precisely; - describe the Kubelka-Munk theory and its limitations as applied in modern computer colour formulation and evaluation of colour strength in dyeing; - apply the appropriate systems and methodologies to perform colour management and colour quality evaluation in the textile and fashion industries and to solve associated problems.
Subject Synopsis/ Indicative Syllabus	(I) The Basic Concepts of Light, Object, And Visual System Interaction of light and object, absorption, reflection, transmission, scattering and their relationship with colour. Interaction of light with visual system, introduction and detection of defective colour visions. (II) Metamerism, Chromatic Adaptation, Colour Rendering of Light Sources Concepts of illuminant metamerism and its detection. Chromatic adaptation, colour rendering properties of light sources and its influence on coloured objects.

Student Study Effort Expected	Class contact:	
	• Lecture	24 Hrs.
	• Laboratory	15 Hrs.
	Other student study effort:	
	• Assignments	30 Hrs.
	• Laboratory report preparation	36 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Essential</u> Berns, R. S. (2000), <i>Billmeyer and Saltzman's Principles of Color Technology</i>. 3rd Edition, John Wiley & Sons. Xin, J. H. (Ed.), (2006), <i>Total Colour Management in Textiles</i>. Woodhead Publishing Ltd, Cambridge, England. Roderick McDonald, (1997), <i>Colour Physics for Industry</i> (2nd Edition), the Society of Dyers and Colourists <u>Supplementary</u> Berger-Schunn, A. (1994), <i>Practical Color Measurement</i>. John Wiley. Fairchild, M. D., (1998), <i>Color Appearance Models</i>. Addison Wesley Longman, Inc. Park, J. (1993), <i>Instrumental Colour Formulation: a practical guide</i>. Society of Dyers and Colourists. <u>Periodicals</u> AATCC review, the American Association of Textile Chemists and Colorists, North Carolina, USA. <i>Color Research and Applications</i>, John Wiley, New York, USA. <i>Coloration Technology</i>, the Society of Dyers and Colourists, United Kingdom.	